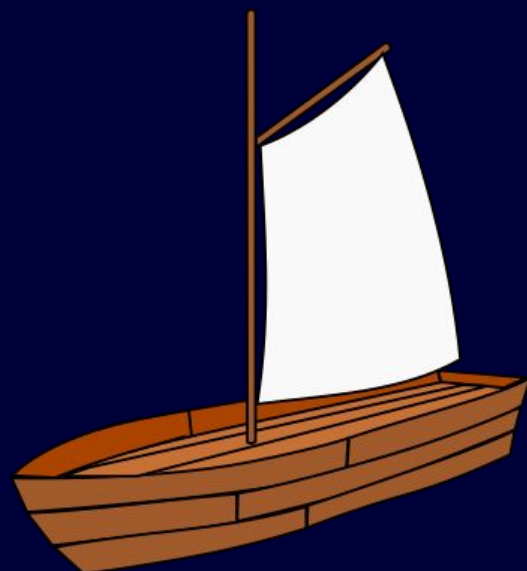
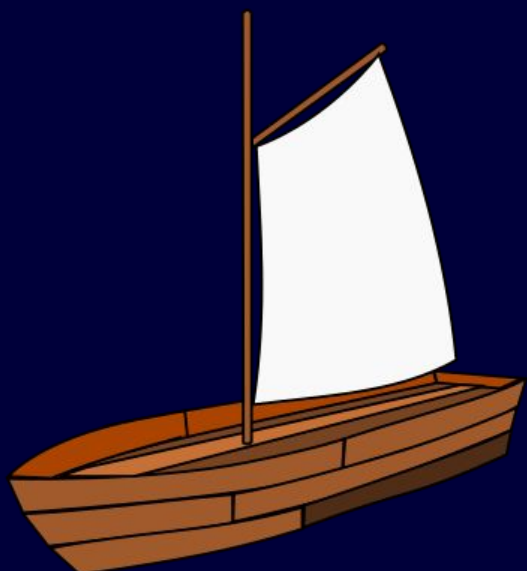
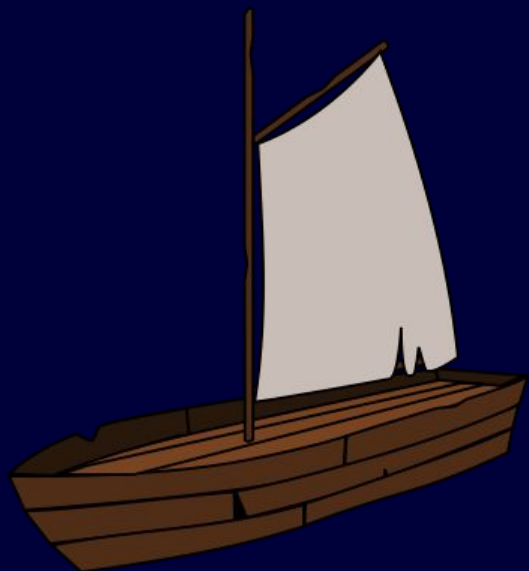


RAG evaluation: A Live Demo

Anupam Krishnamurthy

Continuous Testing Meetup

15.4.2026



Anupam Krishnamurthy

- ❖ MBA turned engineer
- ❖ Head of AI Testing at TestSolutions GmbH
- ❖ Podcast cohost at BeyondQuality
- ❖ Blog: <https://anukrit.de/articles/>

Challenges with LLM Testing

- ❖ Non-determinism
- ❖ Cost
- ❖ Architecture

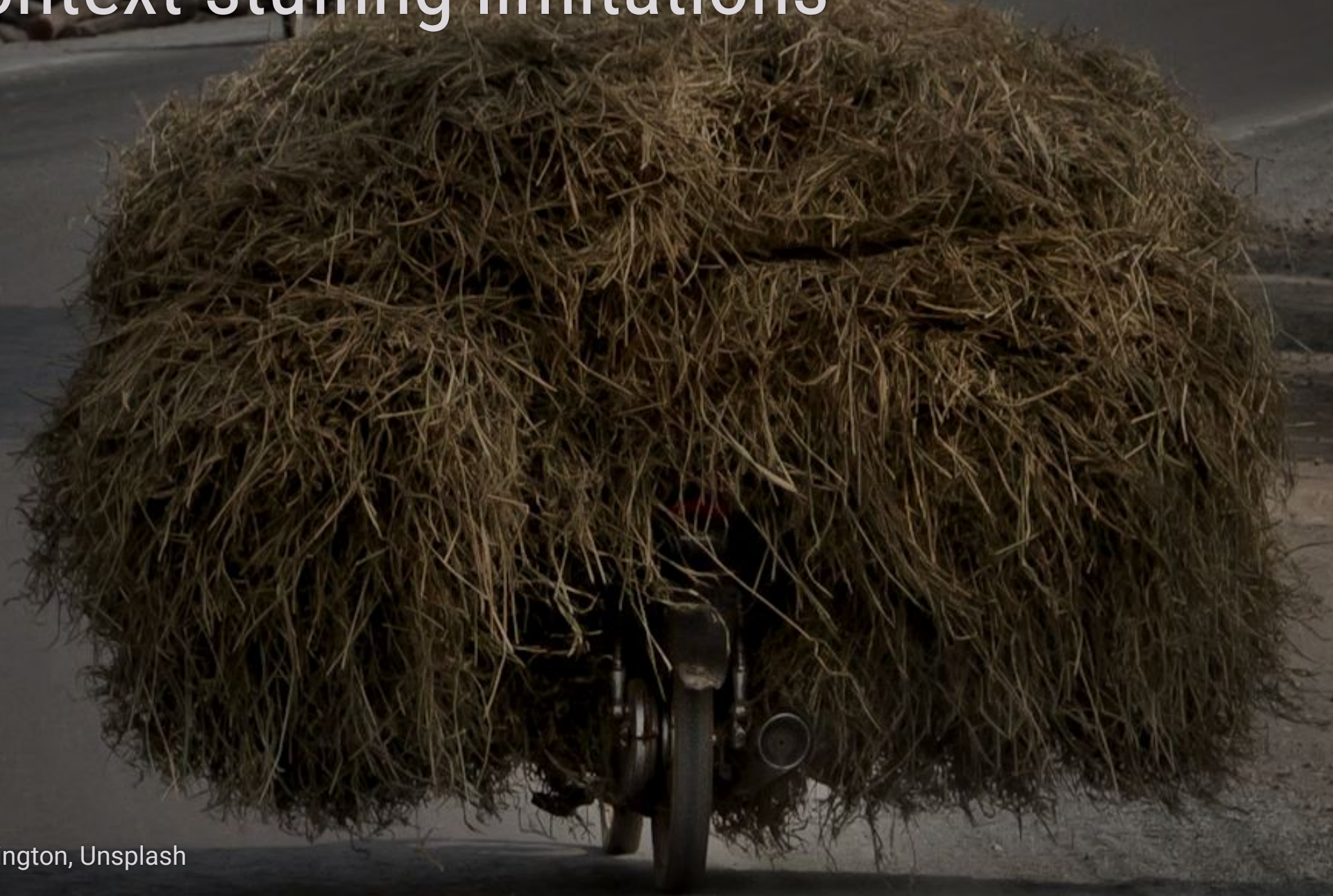
How do we test non-deterministic outputs in an automated manner?

LLM-as-a-judge?



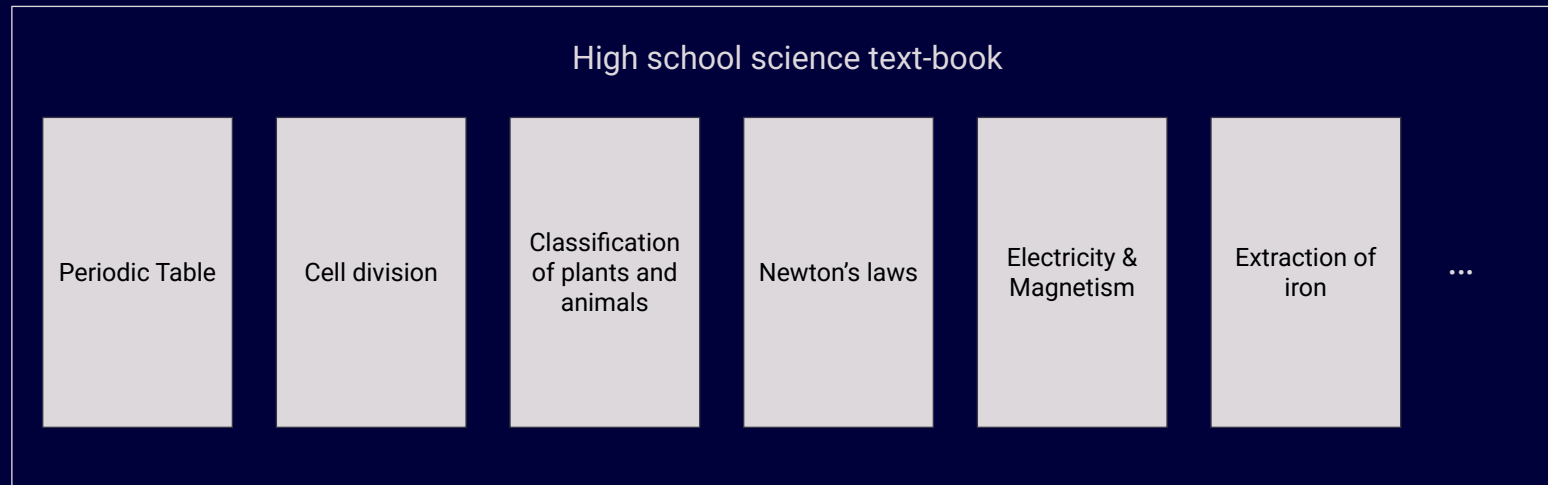
A little bit about Retrieval Augmented Generation

Context stuffing limitations



When we ask an LLM...

We upload a high-school science textbook to an LLM...

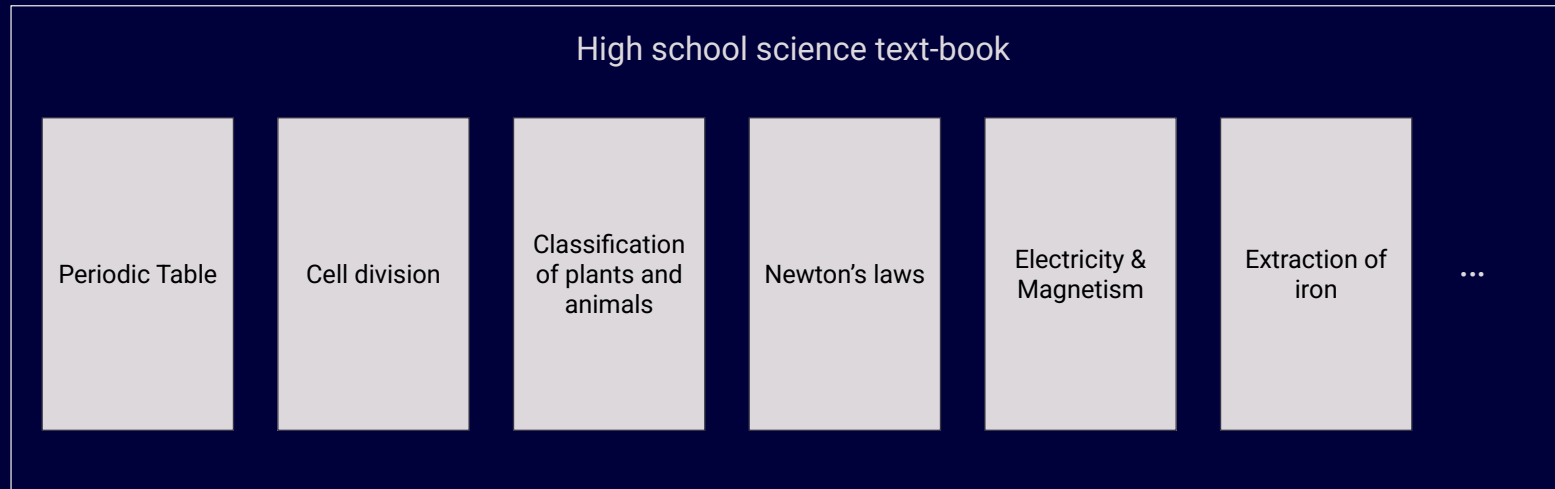


... and ask the LLM: Explain how a rocket works.

Context stuffed LLM

Explain how a rocket works.

LLM context window



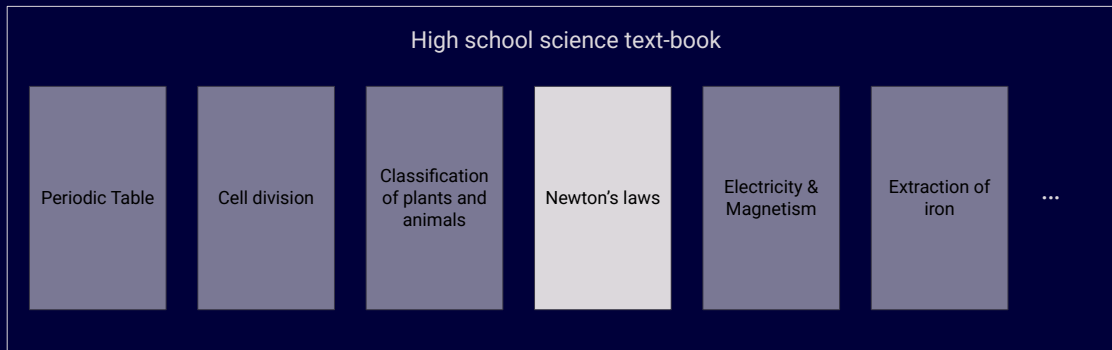
Generation

- 
- Unreliable output
 - Hallucination
 - Higher costs

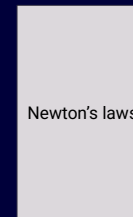
LLM with RAG

Explain how a rocket works.

Retrieval



LLM context window



Generation

- Targeted context
- Reliable output
- Lower costs

Terminology

Embeddings

Vector representations of text

E.g.

"How to reset my password?" $\rightarrow$ [0.11, -0.27, 0.53, ...]

"Change account password" $\rightarrow$ [0.12, -0.25, 0.50, ...]

Cosine similarity $\approx 0.98 \rightarrow$ semantically similar.

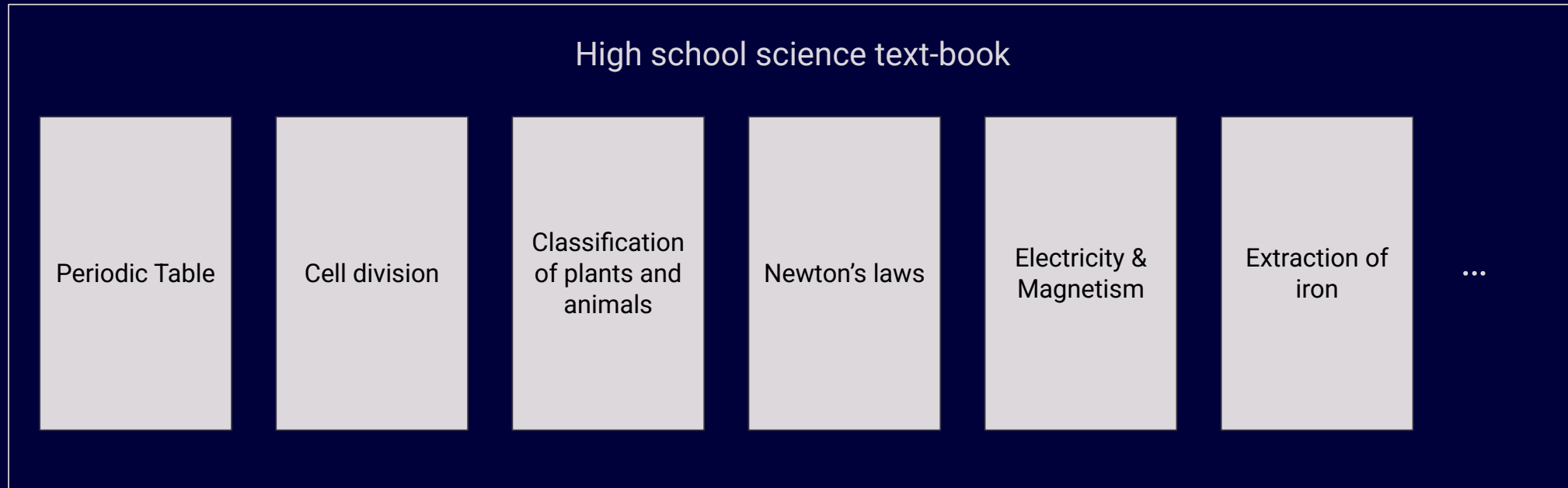
Vector Store / Vector Database

Database optimized for storing embeddings.

Terminology

Chunking

Breaking up document / data-source into smaller pieces



Evals aren't the same as automated tests

- ❖ Non-determinism is the core difference
- ❖ Have thresholds. Not binary pass / fail
- ❖ Are often subjective
- ❖ Regressions are fuzzy
- ❖ Often involve statistical metrics E.g.
 - BERTScore, BLEU, ROUGE (for text similarity)
 - Cohen's Kappa (for inter-rater agreement)
 - Precision, Recall, F1 Score (for classification quality)

Qualifying a RAG model

- ❖ Are all relevant documents retrieved?
- ❖ Are only relevant documents retrieved?
- ❖ Is the answer relevant to the question?
- ❖ Is the answer generated merely from the retrieved context?
- ❖ Is the answer clear and understandable?
- ❖ Does the model say 'I don't know' if relevant context doesn't exist?
- ❖ Is all of the stuff above done consistently?
- ❖ Does the model continue to make sense after 'n' exchanges?
- ❖ Does the model respond fast enough?
- ❖ Does it leak sensitive information when you 'persuade' it?

Demo outline

- ❖ Walk through RAG model setup
- ❖ Explore evaluations with Ragas
- ❖ Debug evaluation with traces using LangSmith
- ❖ Demonstrate security issue

Data source

- ◆ [37 Signals Employee Handbook](#)



Show me. Don't tell me.

What I do differently now

- ❖ Write unit tests for deterministic outputs
- ❖ Start with human judgement for non-deterministic
- ❖ Choose metrics that are closely aligned with judgement
- ❖ Use LLM-as-judge sparingly

How does the quality engineering role change?

- ❖ Return to first principles
- ❖ Grapple with non-determinism
- ❖ Thinking statistically

Try out this demo!

The Beyond Quality community

- ❖ Pursue mastery collaboratively
- ❖ Translate ideas into research or talks
- ❖ Be part of a supportive community+
- ❖ Website: <https://beyondquality.org/>

Links

Source Repo: <https://github.com/anupamck/RAGEvalTalk>

LinkedIn: <https://www.linkedin.com/in/anupam-krishnamurthy/>